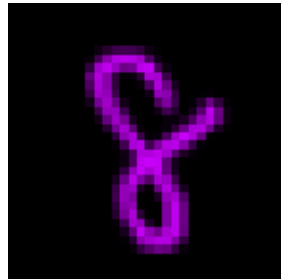
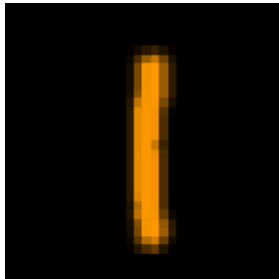
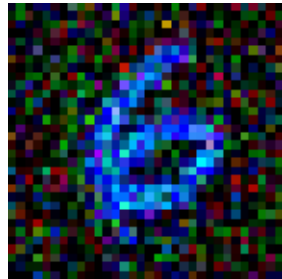
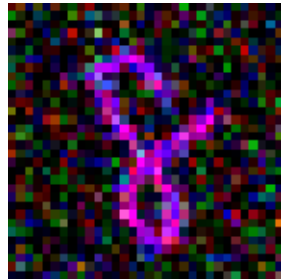
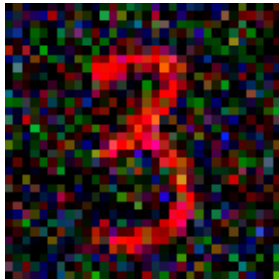
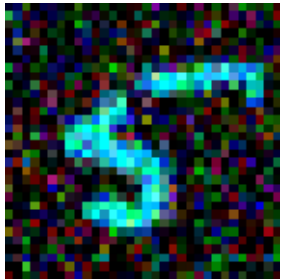
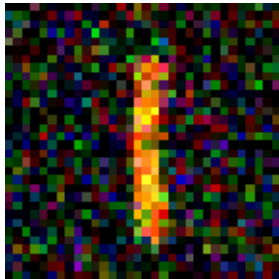


Huber Noise (alpha=1.0, scale=0.5)

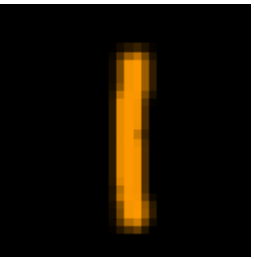
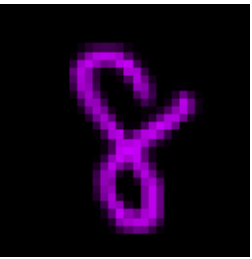
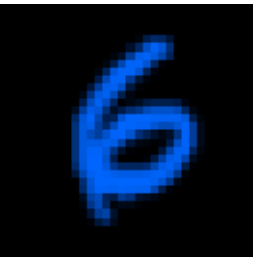
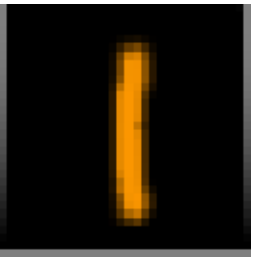
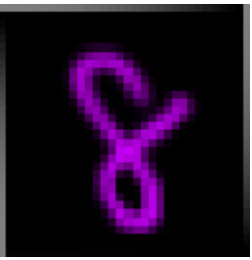
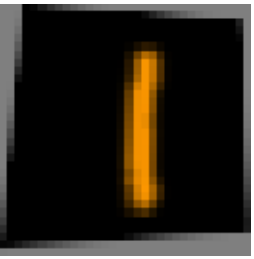
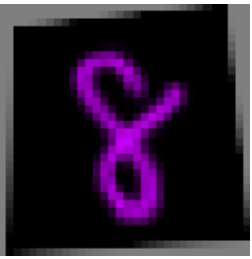
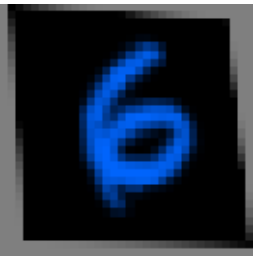
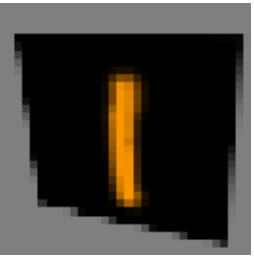
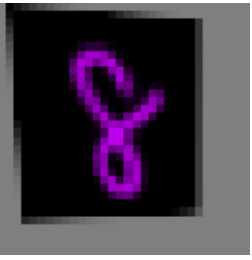
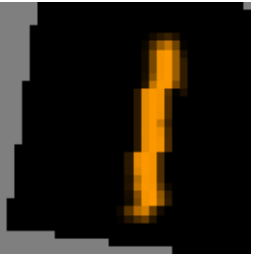
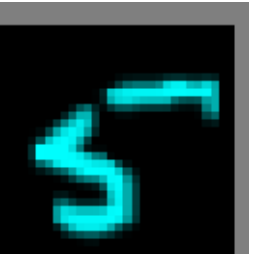
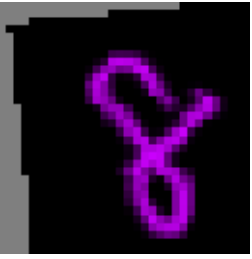
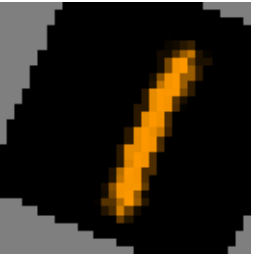
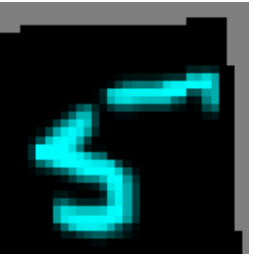
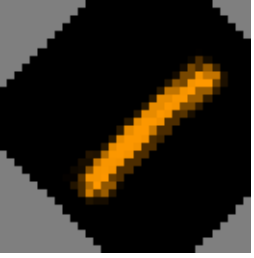
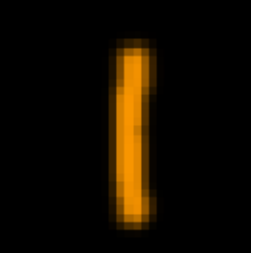
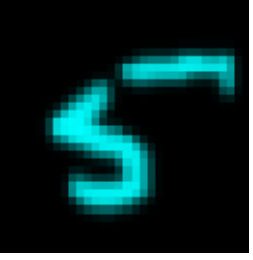
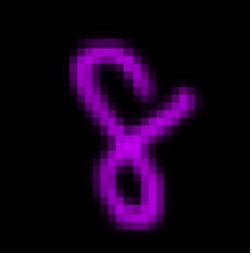
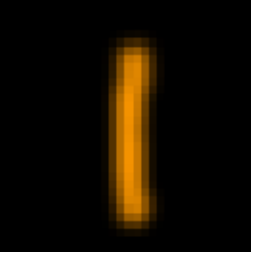
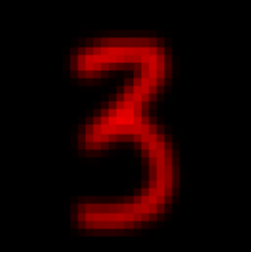
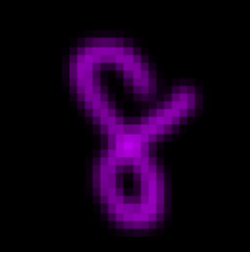
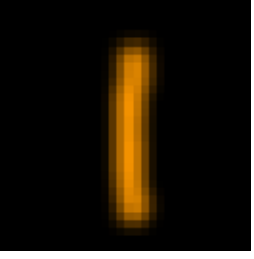
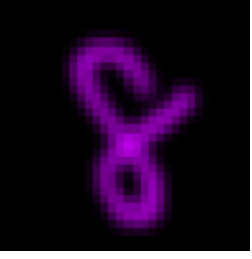
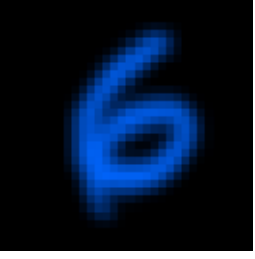
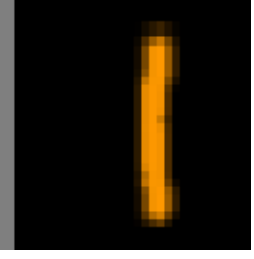
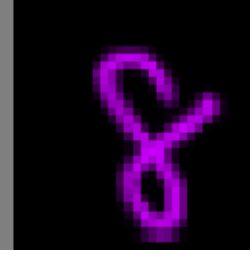
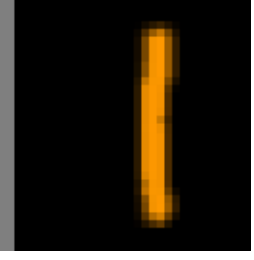
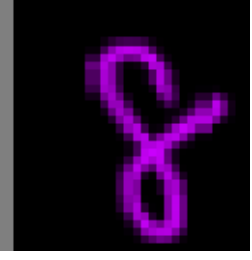
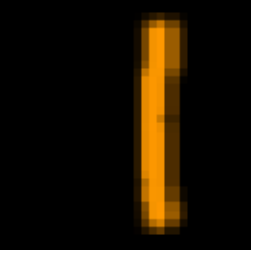
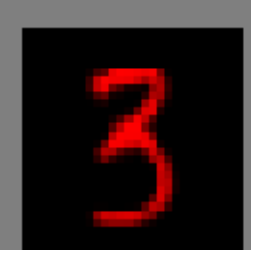
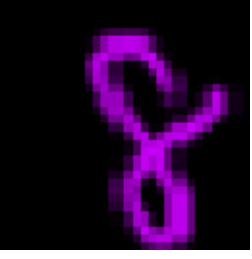
Original



Huber



Camera Shifts Visualization

Original					
Angle (low)					
Angle (med)					
Angle (high)					
Rotation (low)					
Rotation (med)					
Rotation (high)					
Blur (low)					
Blur (med)					
Blur (high)					
Zoom (low)					
Zoom (med)					
Zoom (high)					

Huber Noise - Relative L2 (%)

Method	Alpha=0.0	Alpha=1.0
Abs-LiNGAM (Noisy)	6637.66 ± 945.80 *	7472.17 ± 949.56 *
Abs-LiNGAM (Perfect)	107569.20 ± 13304.38 *	115124.99 ± 13245.34 *
BaryCA	3724666.76 ± 425740.41 *	3770140.32 ± 426951.48 *
DiRoCA (eps_20_delta_0.0)	** 427.53 ± 224.89 **	** 505.54 ± 225.28 **
DiRoCA (eps_40_delta_0.0)	1330.93 ± 391.43 *	1605.11 ± 386.10 *
DiRoCA (eps_61.686502153383906_delta_0.0)	3672.23 ± 1004.72 *	4113.38 ± 991.78 *
DiRoCA (eps_8.0_delta_0.0)	799.27 ± 449.95	897.51 ± 450.42
GradCA	5001759.36 ± 314549.77 *	5064637.39 ± 316695.15 *

** Best Method | * Significantly worse (p<0.05) than best

Camera Shift: Angle - Relative L2 (%)

Method		Low	Med	High
DiRoCA	Abs-LiNGAM (Noisy)	4825.25 ± 541.46 *	3011.88 ± 423.63 *	2081.90 ± 371.23 *
	Abs-LiNGAM (Perfect)	65935.91 ± 4729.22 *	27028.41 ± 1624.55 *	9249.41 ± 920.72 *
	BaryCA	3070447.96 ± 275061.84 *	1112055.76 ± 100551.08 *	278160.11 ± 23932.40 *
	DiRoCA (eps_20_delta_0.0)	** 419.88 ± 236.41 **	** 390.00 ± 211.84 **	** 275.79 ± 111.52 **
	DiRoCA (eps_40_delta_0.0)	1192.56 ± 327.95 *	908.71 ± 187.77 *	521.65 ± 82.71 *
	DiRoCA (eps_61.686502153383906_delta_0.0)	3011.23 ± 772.61 *	1970.65 ± 429.52 *	1043.16 ± 192.93 *
	DiRoCA (eps_8.0_delta_0.0)	720.19 ± 350.00	794.81 ± 346.86	802.83 ± 293.40 *
GradCA		2628923.21 ± 154759.39 *	888630.24 ± 68032.11 *	234984.96 ± 21127.15 *

Camera Shift: Rotation - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		4569.74 ± 574.02 *	4057.84 ± 546.22 *	3698.25 ± 538.91 *
Abs-LiNGAM (Perfect)		63701.61 ± 5498.77 *	50039.09 ± 4736.68 *	41180.43 ± 4116.37 *
BaryCA		2136640.19 ± 205810.58 *	1510796.94 ± 144200.90 *	1088579.02 ± 105443.94 *
DiRoCA	(eps_20_delta_0.0)	** 406.12 ± 220.88 **	** 394.48 ± 227.09 **	** 381.40 ± 230.52 **
	(eps_40_delta_0.0)	1089.78 ± 273.35 *	1010.19 ± 272.11 *	938.99 ± 263.75 *
	(eps_61.686502153383906_delta_0.0)	2669.51 ± 626.03 *	2374.12 ± 565.95 *	2124.83 ± 493.66 *
	(eps_8.0_delta_0.0)	786.92 ± 379.87	774.57 ± 362.78	773.85 ± 341.39
	GradCA	2442041.17 ± 165721.45 *	1831299.09 ± 136757.30 *	1372448.73 ± 109160.25 *

Camera Shift: Blur - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		6610.48 ± 945.59 *	6577.70 ± 945.16 *	6570.57 ± 945.03 *
Abs-LiNGAM (Perfect)		107492.64 ± 13292.10 *	107413.25 ± 13273.12 *	107397.76 ± 13268.12 *
BaryCA		3723919.71 ± 425516.59 *	3722771.51 ± 425179.67 *	3722465.66 ± 425092.15 *
DiRoCA	(eps_20_delta_0.0)	** 426.01 ± 224.29 **	** 424.31 ± 223.37 **	** 423.96 ± 223.12 **
	(eps_40_delta_0.0)	1324.21 ± 390.93 *	1316.42 ± 390.16 *	1314.78 ± 389.96 *
	(eps_61.686502153383906_delta_0.0)	3657.86 ± 1004.39 *	3639.65 ± 1003.76 *	3635.48 ± 1003.56 *
	(eps_8.0_delta_0.0)	796.99 ± 450.11	794.39 ± 450.36	793.87 ± 450.43
	GradCA	5001145.87 ± 314546.35 *	5000186.73 ± 314545.28 *	4999929.10 ± 314545.81 *

Camera Shift: Zoom - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		4879.38 ± 562.23 *	4828.91 ± 567.59 *	4758.84 ± 550.63 *
Abs-LiNGAM (Perfect)		70566.20 ± 5558.73 *	69291.62 ± 6037.77 *	66615.25 ± 6040.89 *
BaryCA		2326585.62 ± 220801.06 *	2304239.81 ± 220816.23 *	2221152.13 ± 219478.69 *
DiRoCA	(eps_20_delta_0.0)	** 406.84 ± 218.01 **	** 403.07 ± 213.17 **	** 383.64 ± 192.87 **
	(eps_40_delta_0.0)	1101.91 ± 273.76 *	1087.93 ± 267.94 *	1030.13 ± 251.37 *
	(eps_61.686502153383906_delta_0.0)	2741.89 ± 639.50 *	2708.46 ± 627.40 *	2568.89 ± 601.17 *
	(eps_8.0_delta_0.0)	792.32 ± 382.14	793.38 ± 381.91	791.58 ± 374.66
	GradCA	2654403.25 ± 179714.77 *	2645577.25 ± 177485.90 *	2667519.18 ± 175004.44 *